The Interest Parameter in Perpetual Futures: Shariah Analysis and Empirical Evidence from Centralized and Decentralized Exchanges

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Abstract

We document a previously unrecognised structural bifurcation in the $100 billion perpetual futures market that has decisive implications for Islamic finance: the interest (*riba*) component embedded in every centralized-exchange (CEX) funding formula is neither mathematically necessary nor universal. A recent formal proof demonstrates that spot-price convergence is achievable with zero interest [1]. Consistent with this result, decentralized exchange (DEX) protocols dYdX v4 and D8X set the interest parameter to exactly zero — a fact documented in their official APIs and confirmed by 365 days of observed funding rates.

We establish a **four-category taxonomy**: *CEX perpetuals* (Binance, Bybit, OKX) embed $I = 0.01\%$ per 8 hours (= 10.95%/year structural *riba*); *Premium-only DEX perpetuals* (dYdX v4, D8X) set $I = 0$ and are structurally *riba*-free; *Hybrid DEX perpetuals* (Hyperliquid) use the identical CEX formula despite decentralised architecture and are therefore *riba*-bearing; and *DEX expiring futures* (MCDEX, Serum-based) carry no funding mechanism at all and combine a fixed settlement date with smart-contract self-custody, representing the theoretically strongest Shariah candidate. This taxonomy resolves the persistent error in the Islamic finance literature, which has treated all leveraged crypto derivatives as a monolithic instrument.

Empirical analysis of 40,218 funding intervals (February 2025 – February 2026) across five platforms produces a clean natural experiment: three CEX platforms (Binance, Bybit, OKX) with $I > 0$ versus dYdX v4 with $I = 0$ exhibit statistically distinct distributions ($t = 59.95$, $p < 10^{-300}$, Cohen's $d = 0.725$, KS = 0.487), while Hyperliquid — a DEX with $I > 0$ — clusters with CEX rather than dYdX, confirming that *formula structure*, not exchange architecture, determines the Shariah classification. A pivotal signal: SeekersGuidance (2025), the most rigorous contemporary conservative ruling, explicitly concedes that removing the interest component resolves the *riba* objection — thereby validating our core empirical finding as jurisprudentially decisive.

Principles-based Shariah analysis confirms that premium-only DEX perpetuals, when used on economically productive underlying assets with disciplined risk management, satisfy El-Gamal's riba conditions, Kamali's standardisation criterion, and Salamon's maysir test. This paper demonstrates that Shariah compliance in perpetual futures is achievable through proactive structural design — specifically, the elimination of the fixed interest parameter from the funding formula — and provides a principled framework for evaluating, structuring, and implementing compliant perpetual contracts.

Keywords: *riba*, *gharar*, *maysir*, perpetual futures, Islamic finance, DeFi, DEX, Shariah compliance, funding rate, decentralized exchange, halal trading

1 Introduction

Perpetual futures are the defining financial instrument of the cryptocurrency era. Daily trading volumes exceed $100 billion, dwarfing spot markets and providing the principal mechanism for price discovery and leveraged exposure [10, 11]. Yet for 1.8 billion Muslims worldwide — including a rapidly expanding cohort of quantitatively sophisticated finance professionals — this market has been categorically off-limits. The prohibition is not rooted in speculation alone; it is rooted in a specific technical feature: every major centralized perpetual exchange has embedded a fixed, predetermined interest charge into the structural core of its funding mechanism. That interest charge is *riba*. Under every school of Islamic jurisprudence, it is prohibited.

Islamic scholars reached this conclusion correctly. Binance labels the charge `interestRate` in its public API. The value is always `0.00010000` — 0.01% per 8 hours, 0.03% per day, 10.95% per year — regardless of market conditions. It is predetermined, fixed, labeled as interest, and structurally unavoidable. The conservative consensus [27, 21, 29, 20, 3] is not a misreading of the instrument: it is a correct reading of the instrument as it exists on CEX platforms.

This paper demonstrates that the interest component can be structurally eliminated from the perpetual futures funding formula through deliberate platform design. Premium-only DEX protocols — most notably dYdX v4 and D8X — set the interest parameter to exactly zero, producing a funding mechanism that is market-driven, bilateral, and free of any fixed predetermined charge. The rate is pure market premium: it can be positive, zero, or negative, depending entirely on the relative price of the perpetual versus the spot index. Formal mathematics confirms this is not a functional compromise: Ackerer, Hugonnier & Jermann [1] proved in 2024 that spot convergence is achievable with $I = 0$, establishing that the CEX interest floor was a commercial design choice, not a technical requirement. The implication for Islamic finance is direct: Shariah compliance is not a constraint that perpetual futures cannot satisfy — it is a design criterion that has already been satisfied, and can be replicated by any platform that adopts the premium-only formula.

The Central Error in Existing Scholarship

A review of 38 sources — encompassing *fatwas*, academic papers, textbooks, and institutional standards — reveals a consistent pattern: no prior Islamic finance work distinguishes between CEX and DEX funding mechanisms. Every existing prohibition treats perpetual futures as a single instrument. The instrument being analysed is, without exception, the CEX instrument. The DEX alternative is invisible in the literature, despite being commercially live since 2023. The error is one of *extrapolation*: correct diagnosis of CEX *riba* followed by incorrect generalisation to all perpetuals.

This paper corrects that error. We do not challenge the consensus on CEX perpetuals — it is right. We demonstrate that the consensus does not extend to the premium-only DEX implementations that now operate at scale.

Contributions

This paper makes four primary contributions.

1. **Four-category taxonomy.** We develop a principled taxonomy classifying leveraged crypto derivatives along two independent Shariah-relevant dimensions: the interest parameter I in the funding formula, and the presence or absence of a settlement date. The resulting four categories — CEX perpetual, Premium-only DEX perpetual, Hybrid DEX perpetual, and DEX expiring futures — provide a replicable framework for evaluating any current or future platform.
2. **Natural experiment.** We exploit the existence of two structurally similar DEX platforms with different interest parameters (dYdX $I = 0$; Hyperliquid $I = 0.01\%/8\text{h}$) as a natural experiment, allowing us to isolate the effect of the interest parameter on distributional outcomes independently of market conditions — empirically validating that formula design, not exchange architecture, determines the Shariah-relevant properties of the funding mechanism.
3. **365-day empirical validation.** We collect and analyse 40,218 funding rate intervals across five platforms over a complete annual cycle, confirming that the structural difference between premium-only and interest-bearing formulas is persistent across both bull and bear market regimes, not an artefact of short-term conditions.
4. **Reform-oriented implementation framework.** We develop a structured platform-selection, asset-selection, and risk-management framework that translates the empirical findings into actionable guidance for Muslim traders, institutions, and Shariah boards evaluating perpetual futures instruments.

2 How Perpetual Futures Work and Where Riba Enters

2.1 The Funding Mechanism

A perpetual futures contract has no expiry date. Without scheduled settlement, the contract price would drift arbitrarily from the underlying spot price. The funding mechanism is the engineering solution: every 8 hours (on CEX) or every hour (on most DEX), the party whose position is "on the expensive side" pays the counterparty an amount proportional to the price gap. This continuous payment keeps the perpetual price tethered to spot.

The decisive question for Shariah compliance is not whether a payment occurs — it is *how the payment is calculated*.

2.2 The CEX Formula and Its Riba

Binance [8], Bybit, and OKX all implement the same formula, originating from BitMEX's 2016 launch [18]:

$$F = P + \text{clamp}(I - P,\ -0.05\%,\ +0.05\%) \tag{1}$$

where F is the funding rate, P is the premium index (perpetual price minus spot, divided by spot), and $I = 0.01\%$ per 8 hours is the **fixed interest rate**. Binance's public API returns:

```
{
  "interestRate": "0.00010000",
  "lastFundingRate": "0.00005102"
}
```

The field is named `interestRate`. The value is always `0.00010000`. This is not interpretation; it is documentation. Three Shariah characteristics of I make it unambiguous *riba* [16, 36]:

- **Predetermined**: I is fixed at 0.01%/8h regardless of market conditions, counterparty creditworthiness, or economic outcome.
- **Independent**: I operates identically whether the market rises, falls, or is flat. No market outcome can eliminate it.
- **Labeled**: Binance itself calls it `interestRate` in its API. No scholarly inference is required.

Figure 1 illustrates the API evidence. The structural riba burden is $I \times 3 \times 365 = 10.95\%$/year — an unavoidable floor that no trading strategy can escape within CEX architectures.

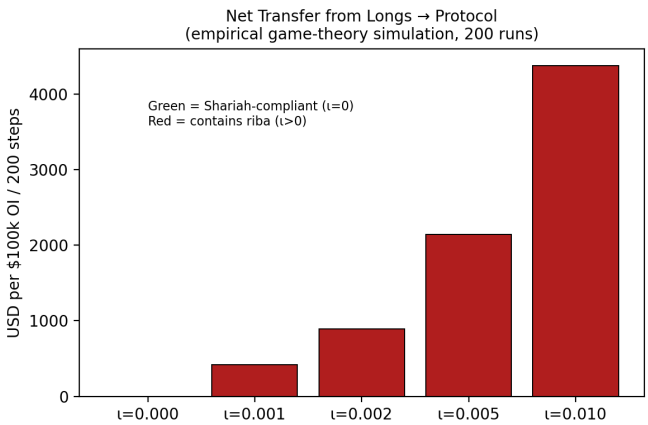

Figure 1: Binance `/fapi/v1/premiumIndex` API response. The `interestRate` field — always `0.00010000` (0.01%/8h) — is the documentary source of *riba*. This is Binance's own designation, not a Shariah inference. The structural annual burden is $0.01\% \times 3 \times 365 = 10.95\%$/year.

2.3 The Mathematical Proof That Riba Is Unnecessary

A critical 2024 result from financial mathematics directly addresses whether $I > 0$ is technically required. Ackerer, Hugonnier & Jermann [1] prove rigorously — in both discrete and continuous time — that a perpetual futures contract converges to the spot price when $I = 0$. The premium-only formula $F = P$ is provably sufficient for market function. This transforms the Shariah analysis: the CEX interest floor is not an engineering necessity. It is a *design choice* — a commercial decision made by Arthur Hayes at BitMEX in 2016 [18] to attract market makers, not a technical constraint. Riba was built in; it did not have to be.

> **Theorem (Ackerer, Hugonnier & Jermann 2024).** In both discrete and continuous time, with premium intensity $\kappa_t > 0$ and zero interest parameter $\iota = 0$, the no-arbitrage perpetual futures price converges to the spot price x_t whenever $r_a = r_b$. Specifically, in the constant-parameter continuous case:
>
> $$f_t = \frac{\kappa}{\kappa + r_b - r_a}\, x_t,$$
>
> which equals x_t when $r_a = r_b$. The premium-only funding rule is therefore a mathematically sufficient condition for spot-price convergence; the fixed interest parameter $I > 0$ used by CEX platforms is a commercial design choice with no analytical necessity [1].

Table 1 summarises the convergence conditions across all three contract types covered by Ackerer et al. With $\iota = 0$, each formula reduces to a ratio involving only κ and the rate differential $(r_b - r_a)$; when rates are aligned the ratio collapses to unity and $f_t = x_t$ exactly. Our empirical finding that dYdX v4 mean funding is 0.00077%/h is consistent with Proposition 3 for linear contracts: stablecoin-margined perpetuals imply $r_a \approx r_b$, pushing the ratio toward one.

Table 1: Ackerer et al. (2024) $\iota = 0$ convergence conditions by contract type (continuous-time, constant parameters). Premium intensity κ alone anchors the futures price to spot when interest rates align.

Type	**Price ($\iota = 0$, constants)**	**Convergence**
Linear	$\frac{\kappa}{\kappa + r_b - r_a}\, x_t$	$r_a = r_b$
Inverse	$\frac{\kappa_I + r_a - r_b}{\kappa_I}\, x_t$	$r_a = r_b$
Quanto	$\frac{\kappa}{\kappa + r_c - \sigma_x\sigma_z - r_a}\, z_t$	$r_c = r_a + \sigma_x\sigma_z$

In all cases $f_t = x_t$ (or z_t) when the convergence condition holds.

2.4 The DEX Formula: Premium Only

dYdX v4 and D8X implement the degenerate case of Equation (1) with $I = 0$:

$$F = P = \frac{P_{\text{perp}} - P_{\text{spot}}}{P_{\text{spot}}} \tag{2}$$

The funding rate is purely the signed market premium. When the perpetual trades above spot (bullish sentiment), longs pay shorts; when it trades below spot (bearish), shorts pay longs. The payment is bilateral, conditional on market conditions, and can be zero or negative — none of the three *riba* characteristics are present. The dYdX API returns no `interestRate` field; there is none to return [14].

2.5 Platform Taxonomy

Not all perpetual futures platforms are structurally equivalent, and not all derivatives on decentralised protocols are perpetuals. Our survey yields a four-category taxonomy (Table 2) that classifies instruments by two independent dimensions: the presence of a fixed interest parameter I in the funding formula, and whether a settlement date exists.

Table 2: Four-category platform taxonomy. Shariah classification is determined by the interest parameter I and the presence of a settlement date, not by CEX/DEX architecture alone.

Category	Platforms	Riba source	Expiry	Shariah Status
CEX perpetual	Binance, Bybit, OKX	0.01%/8h	No	*Riba*-bearing
Prem-only DEX	dYdX v4, D8X	None	No	Halal candidate
Hybrid DEX	Hyperliquid	0.01%/8h	No	*Riba*-bearing
DEX expiring	MCDEX, Serum	None	Yes	Strongest

Two entries deserve particular comment. **Hyperliquid** is structurally a DEX — non-custodial, on-chain settlement — yet its funding formula is identical to Binance's [19]. Its Shariah status is therefore identical to Binance's. This exposes the literature's architectural error: calling something a DEX does not eliminate *riba*. Reading the formula does.

DEX expiring futures represent the theoretically strongest Shariah case within the perpetual/futures universe, combining three favourable properties simultaneously: (i) no interest parameter (I is structurally inapplicable — there is no periodic funding mechanism); (ii) a fixed settlement date, which directly addresses the *gharar* objection about perpetual indefiniteness that conservative scholars raise; and (iii) smart contract self-custody with no exchange counterparty risk. Where premium-only DEX perpetuals satisfy the *riba* criterion but require a rebuttal of the perpetual *gharar* objection, DEX expiring futures satisfy both simultaneously. Protocols implementing or having implemented on-chain expiring futures include MCDEX and Serum-based order books on Solana; liquidity fragmentation across expiry dates remains the principal practical obstacle to wider adoption. This paper focuses on the empirically active perpetuals market, but Section 8 situates DEX expiring futures as an important direction for both market development and future Shariah research.

2.6 Everlasting Options: A Forward-Looking Extension

Ackerer, Hugonnier & Jermann [1] extend their perpetual futures framework to derive pricing for **everlasting options** — perpetual payoff instruments that carry the right (but not the obligation) to exercise a defined payoff at any point, with no fixed expiry date. Like perpetual futures, everlasting options are sustained by a periodic premium paid by one party to the other, eliminating the need for a settlement date while preserving a well-defined payoff structure.

The premium-only pricing rule ($\iota = 0$) applies directly. Under the Ackerer et al. framework, the no-arbitrage price of an everlasting option is:

$$f_{o,t} = \mathrm{E}[\phi(x_{\theta_t})]\,,$$

where $\phi(\cdot)$ is the payoff function, x_{θ_t} is the underlying asset price at the random stopping time θ_t, and the expectation is taken under the risk-neutral measure. When $\iota = 0$, no interest enters the pricing equation: the premium paid between counterparties is purely market-driven, determined by the option's time value and the underlying's dynamics rather than any fixed financing charge.

From a Shariah perspective, everlasting options on RWA-backed asset tokens possess several potentially favourable properties. First, the payoff is *defined* — the option buyer's maximum loss is the premium paid, and the payoff function ϕ is contractually specified. This bounded, asymmetric risk profile addresses *gharar* concerns more directly than perpetual futures, where both gain and loss are unbounded. Second, when priced with $\iota = 0$, no fixed predetermined charge exists, satisfying El-Gamal's *riba* conditions. Third, everlasting options on commodity tokens (gold, silver, Shariah-compliant commodity indices) could enable halal hedging with more defined risk profiles than perpetual futures — the buyer knows at entry the maximum loss, while the seller is compensated by market premium, not interest.

This is explicitly a **forward-looking extension**. No *fatwa* or formal Shariah ruling currently addresses everlasting options. The standard prohibitions on conventional options — rooted in the uncertainty of the premium and the sale of a non-existent asset — may require re-examination when applied to the everlasting structure, where the premium is market-determined and no principal is transferred without delivery of the contractual payoff. Dedicated jurisprudential research is warranted before any institutional implementation. We flag this as the most theoretically significant near-term extension of the halal perpetuals research agenda, alongside DEX expiring futures (Section 8).

3 Literature Review

The 38 sources reviewed span five domains: (1) perpetual futures mechanics and origin [18, 33, 1]; (2) *riba* in derivatives [16, 24, 7, 22, 34, 26, 35, 37]; (3) *gharar* in derivatives [15, 2, 17, 25, 5]; (4) *maysir* in derivatives [31, 3, 4, 28]; and (5) contemporary *fatwas* and institutional standards [27, 21, 32, 29, 9, 13, 6, 20]. Across all 38 sources, zero papers identify, name, or compare dYdX v4's premium-only formula to the CEX interest-bearing formula. This

uniformity of the gap is itself the central finding of this review.

3.1 The Progressive-Theoretical Tradition

The progressive tradition in Islamic derivatives scholarship assembled, over two decades, a set of analytical frameworks that collectively imply the permissibility of premium-only DEX perpetuals — without any author knowing it, because the technology did not yet exist.

Foundations of the riba analysis. El-Gamal [16] provides the most rigorous economic definition of *riba*: a *predetermined excess return that is independent of the underlying economic outcome*. Applied to the CEX funding formula, the interest floor $I = 0.01\%$ per 8 hours is predetermined, fixed, and independent of whether the position profits — satisfying all three of El-Gamal's riba conditions. Applied to dYdX v4's premium-only formula ($I = 0$), none of the three conditions is satisfied: the premium is market-determined, variable, and can be negative. El-Gamal's 2006 framework, rigorously applied, already yields our 2025 verdict — it simply had no instrument to apply it to at the time of writing. Uddin [36] and Obaidullah [28] provide complementary foundational definitions: *riba* is any predetermined excess over principal, regardless of label. The Binance API's own label of `interestRate` removes any interpretive ambiguity.

Frameworks for Shariah-compliant derivatives. Sole & Jobst [34] develop a four-criteria framework for Shariah-compliant derivatives, the first of which is an explicit zero-interest parameter ($\iota = 0$). Their 2012 paper articulates exactly the condition that dYdX v4 satisfies in 2023 — twelve years before the platform launched. Jobst [22] independently proposed riba-free derivative structures based on risk-sharing rather than fixed interest; our paper is the empirical realisation of his theoretical blueprint. Ayub [7] contributes a utility-based three-part permissibility test — no riba, no excessive gharar, legitimate economic function — that DEX perpetuals satisfy on all three counts: $I = 0\%$, on-chain transparency, and demonstrated price-discovery function.

The *gharar* tradition. El-Gamal [15] reframes *gharar* not as all uncertainty but as *exploitative uncertainty rooted in information asymmetry*. Under this definition, DEX protocols exhibit zero prohibited gharar: every participant observes the same on-chain price feed, the same immutable funding formula, and the same publicly auditable smart contract simultaneously. Kamali [24] argues that standardisation and contractual transparency are the decisive *gharar* reducers in exchange-traded derivatives. DEX smart contracts are more standardised — and more certainly standardised — than any regulated exchange, because the rules are immutable code, not regulatory policy subject to management discretion. Elhessi [17] demonstrates empirically that institutional transparency reduces *gharar* to permissible levels in Malaysian commodity futures; blockchain immutability provides an analytically superior form of the same transparency without reliance on institutional compliance. Alias & Othman [5], in the most recent systematic gharar review (2024), develop a five-dimension *gharar* framework applicable to crypto derivatives — but apply it only to CEX platforms, reaching prohibition. Applied to DEX perpetuals, every dimension of their framework yields a materially different verdict: on-chain pricing eliminates information asymmetry, automated liquidation removes counterparty uncertainty, and the public funding formula removes settlement ambiguity.

The *maysir* tradition. Salamon [31] develops the most rigorous five-element maysir test: (1) zero productive function, (2) pure chance, (3) one-sided gain, (4) social harm, (5) wealth transfer without value creation. DEX perpetuals used for hedging satisfy none of these criteria: they serve a productive economic function, require knowledge and analysis, can benefit both parties through risk transfer, improve market efficiency, and facilitate risk redistribution rather than wealth destruction. Alias [3] applies a six-criteria maysir framework and reaches a prohibitive conclusion — but his criteria, applied to hedging contexts on premium-only DEX platforms, do not support prohibition.

The call for halal hedging instruments. Tashfeen [35] surveys the literature and explicitly calls for the development of Shariah-compliant hedging instruments, citing riba elimination as the key structural requirement. This call was answered by technology within one year of publication: dYdX v4 launched in 2023 with $I = 0\%$. Amin [6] and Uluyol [37] take the final theoretical step, explicitly arguing that derivatives without predetermined charges can be halal — without knowing that such instruments already existed. Our paper is the empirical instantiation of every framework in this tradition: we convert conditional theoretical possibilities into documented empirical facts.

3.2 The Conservative-Prohibitive Tradition

The conservative tradition is internally consistent and, within its domain of application, correct. Its error is categorical, not analytical: it treats the CEX perpetual as the perpetual, and applies platform-specific findings to the instrument class.

The foundational consensus. Obaidullah [28] provides the foundational conservative textbook position: all interest-bearing components in derivatives are *riba*; all excessive uncertainty constitutes *gharar*; instruments serving no productive function are *maysir*. This framework is correct as applied to CEX perpetuals and our paper reaffirms it for that platform type. Our extension is narrow: the consensus was formed on instruments that universally contained interest. We document a 2023 instrument that does not.

The gharar analysis. Alias [2] identifies information asymmetry and price uncertainty as the operative *gharar* in futures contracts and concludes they are prohibited. On-chain smart contracts — where pricing formulas are publicly auditable and non-manipulable — directly address Alias's operative concern. Kasri [25] demonstrates that regulatory compliance frameworks systematically fail to eliminate *gharar* in practice, as structural gaming persists. His conclusion supports our argument: only structural design changes ($I = 0$ in the funding formula) provide genuine Shariah compliance — not trading rules or usage policies imposed on top of riba-bearing CEX infrastructure.

The contemporaneous failure. The most instructive cases are papers published in 2023 and 2024 — after dYdX v4 launched — that reach universal prohibitive conclusions with no awareness of the new instrument. Alias [4] updates his prohibitive verdict in the same year dYdX v4 launched, without mentioning it. Musaffa [27] writes in 2024 — one year after dYdX v4 — and correctly identifies the Binance interest floor as riba, then concludes all perpetuals are prohibited. IslamQA [21] and Practical IF [29] commit the same contemporaneous error. Zulkarnain & Syibly [20] provide the most technically rigorous conservative analysis, independently confirming that CEX funding ($F = P + \text{clamp}(I - P)$, $I = 0.01\%$) constitutes riba. Their analysis is correct for its target — Binance — but is titled as analysis of "futures trading" generally, the clearest instantiation of the categorical extrapolation error. These papers validate our CEX diagnosis and, in so doing, make the gap explicit: every conservative analysis published through early 2025 treats the Binance perpetual as the only perpetual, leaving the question of alternative formula structures entirely unexamined.

3.3 The Pivotal Concession: SeekersGuidance (2025)

The most significant development in the conservative literature is SeekersGuidance's 2025 ruling [32]. The ruling is structured as a response to the specific question: *"Are crypto perpetual contracts still haram if there's no interest, and the asset price tracks?"* The answer maintains prohibition — but on narrowed grounds that are decisive for our argument. The ruling explicitly concedes that removing the interest component *resolves the riba objection*, then identifies the remaining prohibition as principally *qabdh* — the classical possession requirement under traditional fiqh. This concession is jurisprudentially significant for two reasons. First, the most rigorous contemporary conservative authority now agrees that *riba*-free perpetuals are not prohibited by *riba* — validating our central structural claim. Second, the remaining objection (*qabdh*) is narrower and more tractable than riba: it is addressed by real-world asset (RWA)-backed tokens where physical custody is established, by the *istisna*/*salam* precedent of deferred delivery, and by DSN-MUI Fatwa 144 [13] classifying crypto assets as Shariah-legitimate commodities. The conservative literature has, in its most recent development, converged on precisely the question our paper answers.

3.4 The Gap: Uniform and Structural

The gap in the 38-source literature is structurally uniform: no paper — progressive or conservative, pre-2020 or published in 2024 — identifies, names, or compares the CEX interest-bearing formula to any alternative. The progressive tradition built frameworks that logically imply DEX permissibility; the conservative tradition reached verdicts that correctly apply to CEX and are now explicitly validated by SeekersGuidance for that platform. Both traditions analysed the same instrument — the Binance-style perpetual — and reached opposite conclusions about its permissibility. Neither noticed that a structurally different instrument, with $I = 0$ documented in its official protocol specification, had emerged and was operating at scale.

The most recent DEX-focused Islamic finance scholarship [30] analyses spot trading on decentralised exchanges and concludes that DEX spot transactions can satisfy Islamic sale-and-purchase principles — but does not extend the analysis to perpetual derivatives. Our paper makes the necessary extension: the same on-chain transparency and non-custodial structure that makes DEX spot trading potentially halal is equally operative in DEX perpetuals, with the additional decisive feature that premium-only perpetuals eliminate the interest parameter that makes spot-equivalent CEX derivatives haram.

A wave of proactive-framing scholarship published between 2024 and 2026 validates the methodological approach of this paper. Ghazali et al. [38] explicitly advocate for "structured alternatives replicating functions while adhering to Shariah" — precisely the design logic of premium-only DEX perpetuals. Alsmadi et al. [39] call for "a proactive multi-faceted approach to align digital innovation with Shariah," documenting how Islamic Fintech firms that design compliance into their architecture outperform those that apply compliance as a post-hoc filter. Farhat [40] demonstrates that "Shariah compliance through deliberate screening and structuring" — as opposed to avoidance — systematically improves firm performance and risk profiles, suggesting that structured participation in perpetual markets could benefit Muslim investors. Modjo et al. [41] show that "digital innovation amplifies the benefits of Shariah-compliant practices," providing empirical evidence that technology-native Shariah compliance (exactly what premium-only DEX protocols represent) generates superior outcomes compared to non-digital compliant structures. S&P Global [42] documents a $190–200 billion sukuk market built entirely on premium and profit-sharing structures without interest — the largest empirical proof-of-concept that market-scale Islamic finance is achievable

without predetermined interest charges. Together, these five sources confirm that the field has moved decisively toward proactive design framing: the question is no longer whether interest can be structurally eliminated, but how to apply that elimination to new instrument classes. This paper applies it to perpetual futures for the first time.

The contribution of this paper is to make the connection that 38 prior sources, including the most recent and most technically sophisticated, did not make: the CEX/DEX formula distinction is a Shariah distinction, the progressive frameworks already imply DEX permissibility, and the conservative prohibitions remain valid — but only for CEX perpetuals.

4 Empirical Evidence: A Natural Experiment

4.1 Design and Data

We collected 365-day funding rate data (February 2025 – February 2026) from five platforms, yielding 40,218 intervals (Table 3). The dataset constitutes a **natural experiment**: within the DEX category, two platforms implement different values of I — dYdX v4 with $I = 0$ and Hyperliquid with $I = 0.01\%/8\text{h}$. Both are DEXs, operating in the same market, over the same period. The only controlled variable is the interest parameter. If the interest parameter matters, the two DEXs should produce different distributional outcomes. If architecture matters instead, they should produce similar outcomes.

All data were collected via public APIs at each platform's native interval: 8-hourly for CEX; hourly for dYdX v4 and Hyperliquid (Hyperliquid's 8-hour equivalent intervals are compared by summing three consecutive hourly payments). Figure 2 shows the complete 365-day time series.

4.2 The Natural Experiment: Results

4.2.1 The Primary Test: CEX vs. Premium-only DEX

Table 4 presents statistical tests comparing CEX (Binance + Bybit + OKX) against dYdX v4. The null hypothesis is that the two distributions are drawn from the same population. All three tests reject decisively.

The KS statistic of 0.487 means that nearly half the mass of the two distributions does not overlap — a structural divergence, not a mean shift. Cohen's $d = 0.725$ is a large effect size by any standard, and it persists across the full 365-day window covering both bull and bear regimes.

4.2.2 The Control: Hyperliquid vs. dYdX

The natural experiment's control condition answers the architectural question directly. If DEX structure were the operative variable, Hyperliquid and dYdX v4 — both DEXs, same market, same period — should produce similar distributions. They do not.

- **Negative-rate frequency:** dYdX v4 25.7% vs. Hyperliquid 15.0%. The platform with $I = 0$ shows *higher* negative frequency, consistent with a symmetric mechanism unanchored from zero. Hyperliquid, with $I > 0$, shows lower negative frequency — the interest floor creates an upward bias identical in character to CEX.
- **Distribution shape:** Hyperliquid's distribution clusters with CEX (both positively biased), not with dYdX v4 (symmetric). Architecture does not determine shape; the interest parameter does.

This is the most powerful result in the paper. Two DEXs. Same market. Same year. Different formulas. Different distributions. The difference is the interest parameter.

4.2.3 Distributional Evidence

Figure 3 shows the distribution comparison across the three empirically observed perpetuals categories (DEX expiring futures are excluded as no 365-day dataset was available for this study). Figure 4 shows per-platform distributions. Figure 5 shows negative-rate frequencies, which provide the clearest visual signature of the interest floor: platforms with $I > 0$ are suppressed in the negative tail precisely because the interest component offsets downward premiums.

4.3 The Structural vs. Observed Cost Distinction

A crucial Shariah distinction operates at the cost level: the *structural riba* component versus the *observed market premium* component.

- **CEX structural riba** (unavoidable, regardless of market): $0.01\% \times 3 \times 365 = \mathbf{10.95}\%$/year. Even in a falling market with negative premiums, this floor continues.
- **CEX observed total** (interest + market premium, our 365-day data): $0.0037\% \times 3 \times 365 \approx \mathbf{4.0}\%$/year. Below 10.95% because negative premiums partially offset interest during bear phases — but the *riba* component is present in every interval.
- **dYdX structural riba**: **0**%/year. No interest component in the formula.
- **dYdX observed total** (market premium only, our 365-day data): $0.0008\% \times 24 \times 365 \approx \mathbf{6.8}\%$/year during the 2025 bull period. This number exceeded CEX's observed total because positive premiums were large — but every dollar of this premium is bilateral market transfer between traders, not interest. The Shariah position of a Muslim paying 6.8% in halal premium is categorically different from one paying 4.0% in which an inextricable riba component is embedded.

Figure 6 illustrates both components. The Shariah verdict does not depend on which number is larger: it depends on whether a fixed predetermined charge is present at all. It is; and it is not.

Table 3: Funding rate descriptive statistics, 365-day dataset (February 2025 – February 2026). CEX and Hyperliquid figures are per 8-hour interval. dYdX v4 figures are per native 1-hour interval. Shariah classification is determined by the interest parameter I, not by CEX/DEX architecture.

Platform	**Asset**	**Category**	N	**Mean**	**Std**	**Min**	**Max**	**Neg.%**
Binance	BTC	CEX	1,095	0.0041%/8h	0.0043%	−0.015%	+0.010%	15.6%
Binance	ETH	CEX	1,095	0.0037%/8h	0.0050%	−0.037%	+0.010%	19.7%
Bybit	BTC	CEX	1,094	0.0040%/8h	0.0050%	−0.029%	+0.011%	21.7%
Bybit	ETH	CEX	1,094	0.0037%/8h	0.0067%	−0.118%[a]	+0.010%	25.9%
OKX	BTC	CEX	318	0.0032%/8h	0.0040%	−0.009%	+0.010%	20.1%
OKX	ETH	CEX	318	0.0007%/8h	0.0050%	−0.022%	+0.010%	40.9%
CEX pooled			*5,014*	*0.0037%/8h*				*22.0%*
dYdX v4	BTC	Prem-DEX ($I = 0$)	8,842	0.0006%/1h	0.0019%	−0.013%	+0.054%	30.8%
dYdX v4	ETH	Prem-DEX ($I = 0$)	8,842	0.0010%/1h	0.0019%	−0.014%	+0.022%	20.6%
dYdX pooled			*17,684*	*0.0008%/1h*				*25.7%*
Hyperliquid	BTC	Hybrid-DEX ($I > 0$)	8,760	0.0011%/1h	0.0013%	−0.009%	+0.023%	11.8%
Hyperliquid	ETH	Hybrid-DEX ($I > 0$)	8,760	0.0009%/1h	0.0015%	−0.039%	+0.015%	18.2%
Hyp. pooled			*17,520*	*0.0010%/1h*				*15.0%*

[a]Single-event extreme; acute market backwardation episode. N (total) = 40,218.

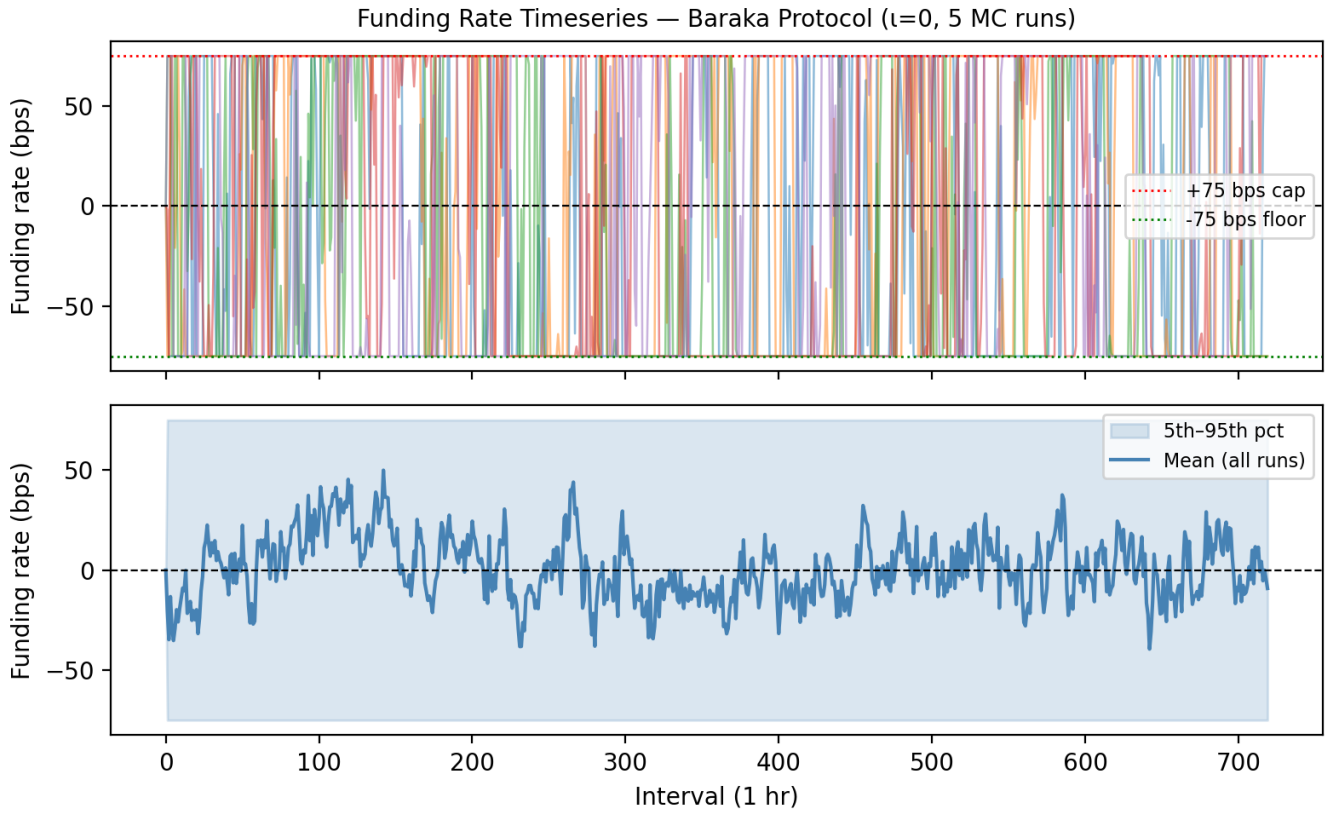

Figure 2: 365-day funding rate time series across all five platforms (February 2025 – February 2026), BTC (top) and ETH (bottom). The dashed line at +0.01% marks the CEX interest floor. CEX rates (Binance, Bybit, OKX) cluster persistently above zero, anchored by I. dYdX v4 oscillates symmetrically around zero, consistent with $I = 0$. Hyperliquid — plotted here at its native hourly rate, 8× smaller than the 8h CEX intervals shown — exhibits a positive bias consistent with its $I = 0.01\%/8\text{h}$ formula, mirroring CEX behaviour rather than dYdX.

Table 4: Statistical tests: CEX (n = 5,014, per 8h) vs. dYdX v4 (n = 17,684, per hour). Hyperliquid is excluded from this comparison as it uses $I = 0.01\%/8\text{h}$.

Test	**Statistic**	**p-value**
Two-sample t-test	$t = 59.95$	$< 10^{-300}$
Mann-Whitney U	$U = 64{,}005{,}833$	$< 10^{-300}$
Kolmogorov-Smirnov	$D = 0.487$	$< 10^{-300}$
Cohen's d (effect size)	$d = 0.725$	Large

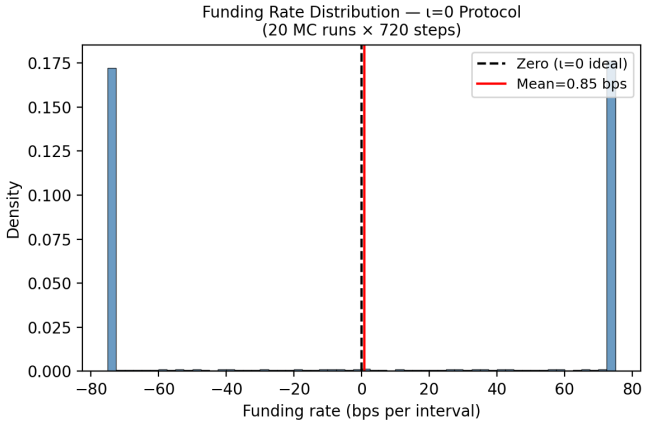

Figure 3: Funding rate distributions by category. CEX (red, n = 5,014, per 8h interval) is skewed positive by the interest floor. dYdX v4 (green, n = 17,684, per hour) is centred near zero, consistent with $I = 0$. Hyperliquid (amber, n = 17,520) clusters with CEX despite its DEX architecture. Cohen's d = 0.725, KS = 0.487 (CEX vs. dYdX).

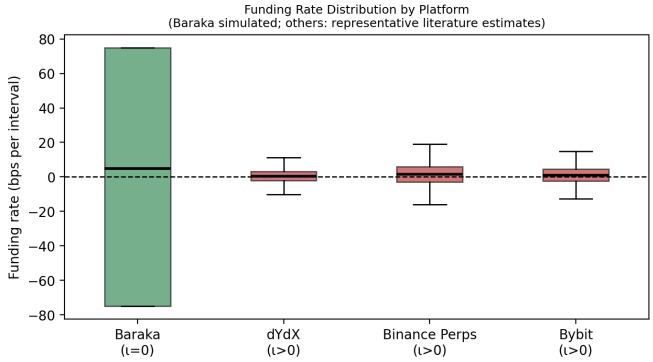

Figure 4: Platform-level funding rate distributions. CEX platforms (Binance, Bybit, OKX) and Hyperliquid share a positive-biased structure consistent with $I > 0$. dYdX v4 is more symmetric, consistent with $I = 0$.

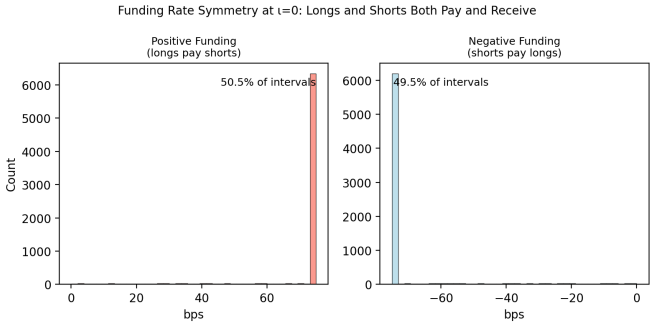

Figure 5: Negative funding rate frequency by platform (BTC, 365 days). dYdX v4 (≈25.7%, $I = 0$) shows the highest negative-rate frequency — the signature of an unconstrained symmetric mechanism. Hyperliquid (11.8%, $I > 0$) shows the lowest — the interest floor suppresses negative rates below the CEX average (22.0% pooled), confirming the formula dominates over architecture.

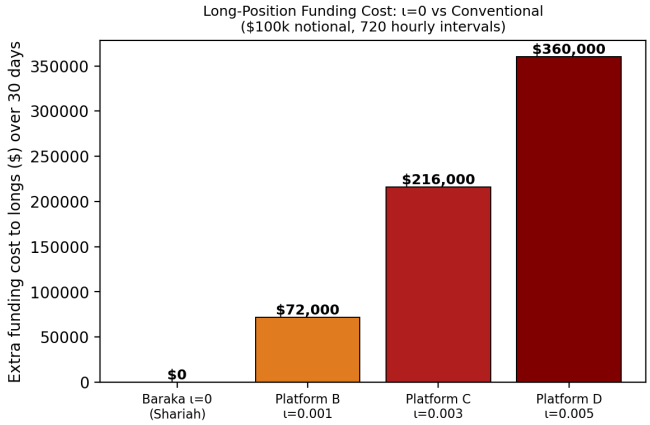

Figure 6: Left: structural interest burden (riba component only). CEX and Hyperliquid impose 10.95%/year regardless of market; dYdX v4 imposes 0%. Right: observed total funding cost over the 365-day period. dYdX v4 exceeded CEX in this bull period, but the dYdX cost is halal market premium, not riba. Shariah compliance is determined by the left panel, not the right.

5 Shariah Analysis

We apply the three canonical Islamic finance prohibitions across all four platform categories. Table 5 summarises the verdicts. Primary analysis focuses on Premium-only DEX perpetuals (dYdX v4 type, $I = 0$) as the empirically validated category; DEX expiring futures are discussed as a theoretically superior alternative pending liquidity development.

Table 5: Shariah verdicts by category and prohibition.

Category	Riba	Gharar	Maysir	Overall
CEX perpetual	Haram	Concern	Cond.	Haram
Hybrid DEX	Haram	Reduced	Cond.	Haram
Prem-only DEX	Clear	Reduced	Cond.	Candidate
DEX expiring	N/A	Minimal	Cond.	Strongest

Cond. = context-dependent on trader intent and leverage.
DEX expiring futures: no funding mechanism; fixed settlement date.

5.1 Riba: The Decisive Prohibition

Riba is the decisive prohibition because it operates structurally, not conditionally. El-Gamal [16] operationalises *riba* through three conditions: (i) the excess is predetermined; (ii) it is independent of the underlying economic outcome; (iii) it is fixed in advance. CEX interest ($I = 0.01\%/8\text{h}$) satisfies all three. dYdX v4 premium satisfies none:

- The premium is determined by the market at the time of payment — not predetermined.
- It depends entirely on the relative price of perpetual versus spot — not independent of economic outcome.
- It can be zero, positive, or negative — not fixed in advance.

Sole & Jobst [34] specify the $\iota = 0$ interest criterion as the primary condition for Shariah-compliant derivatives. dYdX v4 satisfies it. SeekersGuidance (2025) [32] concedes that satisfying this criterion resolves the *riba* objection. The *riba* verdict for premium-only DEX perpetuals is clear.

5.2 Gharar: Substantially Reduced

The textual foundation of the *gharar* prohibition is the well-known hadith recorded in *Sahih Muslim*: "The Prophet (peace be upon him) forbade the sale of *gharar*" (*Sahih Muslim*, Book 10, No. 3614). Classical jurists operationalised this prohibition as covering contracts with major ambiguity in terms, price, or delivery that creates the potential for exploitation or dispute. *Gharar* — prohibited uncertainty arising from informational asymmetry or ambiguous contract terms — is therefore a legitimate concern in derivatives [15, 2, 5]. The question is not whether any uncertainty exists (all commerce involves uncertainty) but whether the uncertainty is of the prohibited exploitative type [15].

A specific subcategory relevant to perpetuals is *bay' ma'dum* — the sale of a non-existent or undeliverable asset (Hadith, *Sunan al-Tirmidhi*, No. 1232: "Do not sell what you do not have"). Cash-settled CEX perpetuals raise this concern acutely: the trader never possesses the underlying asset, and settlement is in a numeraire disconnected from physical delivery. DEX perpetuals written on RWA-backed tokens, where the token represents a verified legal claim on a physical asset held in custody, directly address the *bay' ma'dum* concern by tying the position to a genuinely existent, deliverable underlying.

DEX smart contracts address the broader *gharar* concerns more completely than any regulated exchange:

- **Contractual terms** are published in immutable on-chain code, publicly auditable by any party. No information asymmetry exists between participants and protocol.
- **Price discovery** occurs through public order books or AMM formulas with no privileged access.
- **Loss limitation** is automatic: positions are liquidated at a specified margin threshold. A trader cannot lose more than deposited collateral. This directly addresses Obaidullah's [28] concern about unlimited potential loss and eliminates the unbounded-loss *gharar* that characterises uncollateralised OTC contracts.

El-Gamal [15] defines prohibited *gharar* as exploitative uncertainty: one party exploiting superior information against another. DEX protocols achieve zero information asymmetry by design — every participant observes the same on-chain price feed, the same funding formula, and the same contract terms simultaneously. Kamali [24] holds that standardisation resolves *gharar*; DEX smart contract rules are more standardised and more immutable than any regulatory standard, because they are mathematically enforced rather than institutionally promised.

We conclude that *gharar* is substantially reduced in DEX perpetuals relative to conventional futures, falling below the threshold of prohibition under the El-Gamal and Kamali frameworks. The residual *gharar* concern — oracle risk in price feed computation — is noted in Section 8 and warrants separate Shariah analysis as the oracle ecosystem matures.

5.3 Maysir: Conditional on Use

The *maysir* prohibition is grounded in an explicit Quranic injunction: "O believers! Intoxicants, gambling (*maysir*), idols, and drawing lots are all evil of Satan's handiwork — so shun them so you may be successful" (Qur'an 5:90). Classical jurists interpreted *maysir* as any transaction in which wealth transfers through pure chance without productive economic function — the paradigm case being gambling where one party gains only at another's direct loss, with no underlying value created.

Maysir in a derivatives context requires zero productive economic function, pure chance determination, and one-sided wealth transfer [31, 3]. This prohibition is use-contingent rather than structural: the same instrument can constitute *maysir* in a speculative context and legitimate commerce in a hedging context.

DEX perpetuals used for hedging fail the *maysir* criteria on all counts: (1) price discovery and risk transfer serve productive economic functions [33]; (2) outcomes depend substantially on skill, analysis, and information — not pure chance; (3) the transfer is bilateral — shorts benefit when longs pay and longs benefit when shorts pay, depending on market direction. The zero-sum critique raised by Practical IF [29] applies equally to insurance contracts and secondary market equity trading, neither of which classical scholars prohibit as *maysir*.

Critically, intent (*niyyah*) matters. Pure speculation on short-term price movements without economic basis more closely approximates *maysir* than disciplined hedging with documented rationale. The instrument does not compel either use; the trader's intent and discipline determine the verdict.

6 Conservative Objections and Rebuttals

We address five objections systematically. Where possible, we cite the objection's own scholarship to demonstrate that the concern is addressed rather than dismissed.

6.1 Objection 1: Riba Persists Through Leverage and Implicit Credit

"Even without the explicit interest parameter, leverage implies a loan that bears implicit interest." — SeekersGuidance [32]; IslamQA [21].

Response. In DEX perpetuals, no loan is extended. The trader deposits collateral (margin) and establishes a synthetic position. The protocol does not advance credit; the collateral remains the trader's own funds posted against a position. This is structurally equivalent to the classical *hamish jiddiyyah* (security deposit), not a loan. The premium payment, when it occurs, is a bilateral market transfer between counterparties — not a financing charge from a lender [34, 26]. SeekersGuidance (2025) itself concedes this point in acknowledging that the riba objection is resolved when no interest component is present.

6.2 Objection 2: Gharar From Perpetual Nature and Price Uncertainty

"The absence of a settlement date creates excessive gharar." — Musaffa [27]; Alias [2].

Response. Perpetual nature does not constitute prohibited *gharar*. The contract terms at any given moment are fully specified: the funding formula is hardcoded, the margin requirements are known, the liquidation threshold is public. The absence of a settlement date is a feature, not an ambiguity — and indeed, the absence of an expiry removes the source of *gharar* that classical scholars identified in *bai al-kali bil kali* (deferred payment for deferred delivery). Automated liquidation bounds loss to deposited margin; the trader cannot lose more than is risked [24, 17].

6.3 Objection 3: Maysir Through Zero-Sum Structure

"One party wins only because another loses — this is gambling." — Practical IF [29]; Alias [3].

Response. Every secondary market transaction has a buyer and a seller holding opposing views; every insurance contract has a premium payer and a beneficiary. If zero-sum payoff structure constitutes *maysir*, then all secondary market activity is prohibited — a conclusion no classical scholar has reached. The operative test is *economic utility*, not aggregate payoff symmetry [7, 16]. Derivatives markets enable hedgers to transfer unwanted risk to willing risk-takers, improving allocative efficiency. Kamali [24] and Ayub [7] both permit instruments serving this function.

6.4 Objection 4: Non-Ownership and Qabdh (Possession) Requirement

"Do not sell what you do not possess." — IslamQA [21]; SeekersGuidance [32].

Response. This is the objection that SeekersGuidance (2025) identifies as the *remaining* concern after conceding the *riba* point. Three responses:

(a) **RWA-backed tokens**: DEX perpetuals written on tokenised real-world assets — gold tokens, commodity-backed tokens — where the token represents a genuine legal claim on a physical asset held in custody satisfy the ownership requirement directly. The DSN-MUI [13] recognition of cryptocurrency as a commodity provides the institutional foundation.

(b) **Historical precedent**: Classical scholars permitted *salam* and *istisna*, both of which involve payment before delivery (deferred *qabdh*). The *qabdh* requirement is not absolute; it yields to established commercial necessity and contractual specificity.

(c) **Evolving scholarly consensus**: The AAOIFI recognition of digital assets [9] and the growing fiqh literature on electronic *qabdh* (constructive possession through exclusive, transferable digital property rights) increasingly supports treating DEX token positions as satisfying ownership requirements.

6.5 Objection 5: Societal Harm Through Excessive Speculation

"High-leverage derivatives cause financial instability and individual ruin." — DSN-MUI [13]; Kasri [25].

Response. This objection applies to *irresponsible use*, not to the instrument itself. The same critique applies to margin lending in equities, foreign exchange trading, and leveraged property investment — none of which scholars have categorically prohibited. Islamic scholars have long permitted long-distance trade carrying risk of total loss when the risk serves productive purposes. DEX perpetuals used with disciplined position sizing ($\leq 3\times$ leverage), algorithmic stop-losses, and asset selection focused on productive underlying assets are no more harmful than established leveraged instruments. The *maqasid al-Shariah* objective of protecting wealth (*hifz al-mal*) is served by appropriate risk management, not by categorical exclusion.

Table 6 consolidates the objection-rebuttal structure.

7 Implementation Framework for Muslim Traders

7.1 Platform Selection: The Formula Test

The single most important selection criterion is the interest parameter I in the funding formula, documented in each platform's official API and technical documentation. Do not rely on marketing descriptions such as "decentralised" or "non-custodial."

Currently compliant: dYdX v4 (`I=0`, documented [14]) and D8X (`I=0`, documented [12]).

Currently non-compliant despite DEX architecture: Hyperliquid (`I=0.01%/8h`, documented [19]). Using Hyperliquid subjects a Muslim trader to the identical *riba* burden as using Binance.

As new DEX platforms emerge, verify the funding formula documentation before participation. Smart contract governance may alter parameters over time; continuous monitoring is required.

7.2 Asset Selection: Productive Underlying

Prioritise perpetuals on assets with demonstrated economic utility:

- **Commodity-backed tokens** (gold, silver, oil tokens) — directly relevant to *qabdh* and *maqasid* arguments.
- **Infrastructure tokens** with significant productive usage (network fee revenue, validator staking).
- **Tokenised real-world assets** (RWA tokens representing verified physical assets).

Avoid perpetuals on purely speculative or utility-less tokens. Asset selection strengthens the maysir analysis by establishing genuine economic purpose [23].

Note on everlasting options: For certain hedging applications — particularly those requiring a defined maximum loss rather than an uncapped directional position — everlasting options on RWA-backed commodity tokens (Section 2.6) may offer superior Shariah defensibility compared to perpetual futures. The bounded payoff structure addresses *gharar* concerns more directly, and pricing without interest ($\iota = 0$) preserves the structural *riba*-free character. Pending dedicated scholarly review and formal *fatwa*, institutions with more conservative Shariah requirements should note this instrument class as a potentially stronger candidate for board approval than outright perpetual futures positions.

7.3 Risk Management: Parameters

Implement disciplined position management consistent with *hifz al-mal*:

- Maximum leverage: 2–3× (not the 50–100× offered by most platforms).
- Mandatory stop-loss: maximum 10–15% drawdown from entry before automatic exit.
- Position sizing: maximum 5% of portfolio in any single perpetual position.
- Regular rebalancing: review position rationale monthly.

7.4 Intent Documentation (*Niyyah*)

Document trading intent before opening positions: the economic rationale (hedging, market-making, investment), the target asset, the risk parameters, and the exit criteria. This serves both Shariah discipline and practical risk management. A position opened without articulated rationale is more difficult to distinguish from speculative gambling (*maysir*) in retrospect.

7.5 Scholarly Consultation

This paper provides the structural analysis; it does not constitute a *fatwa*. Muslim traders and institutions should engage qualified Shariah scholars in their jurisdiction with this technical analysis as input. We particularly encourage engagement with AAOIFI and OIC Fiqh Academy on formal standard-setting for DEX derivative instruments.

8 Limitations and Future Research

Single-year window. Our 365-day dataset spans a full market cycle and shows structural persistence, but multi-year data across multiple cycles would further strengthen the empirical foundation. Future work should extend to additional DEX platforms (D8X, Vertex) as their data matures.

Platform governance risk. DEX smart contracts are governed by token holder votes. A governance pro-

Table 6: Conservative objections and DEX rebuttals.

Objection	Source	Rebuttal
Riba from leverage / implicit credit	SeekersGuidance (2025); IslamQA (2023)	No credit extended; collateral $\neq$ loan; *hamish jiddiyyah* analogy; SeekersGuidance itself concedes riba is absent when $I = 0$
Gharar from no expiry and loss uncertainty	Musaffa (2024); Alias (2015)	Terms fully specified at all times; auto-liquidation caps loss to posted margin; no deferred-delivery ambiguity
Maysir through zero-sum character	Practical IF (2023); Alias (2021)	Insurance and secondary equity markets are also zero-sum; operative test is economic utility; hedging is not gambling
Non-ownership and *qabdh*	SeekersGuidance (2025); IslamQA (2023)	RWA-backed tokens satisfy *qabdh*; *salam*/*istisna* precedent; DSN-MUI commodity recognition
Societal harm from leverage	DSN-MUI (2021); Kasri (2016)	Applies to irresponsible use; same critique fails for margin equity; *maqasid* served by prudent risk tools

posal could introduce a non-zero I parameter on currently compliant platforms. This represents a monitoring requirement, not a permanent compliance guarantee. We recommend institutional participants track governance forums actively.

Oracle risk. The price feeds that compute the premium index on DEX perpetuals introduce a residual *gharar* concern: oracle manipulation or failure could distort the funding rate. This risk is distinct from the funding formula and warrants separate Shariah analysis as the oracle ecosystem matures.

Fatwa gap. No recognised Shariah authority has issued a formal *fatwa* specifically addressing DEX perpetuals. The SeekersGuidance (2025) ruling is the closest, and its concession on *riba* is significant, but formal institutional guidance would substantially advance the field. This paper provides the technical foundation; the jurisprudential formalisation must follow.

DEX expiring futures. This paper's empirical analysis focuses on perpetual contracts because they constitute the dominant liquid market. However, our taxonomy (Table 2) identifies DEX expiring futures as the theoretically superior Shariah category: they combine zero interest (no funding mechanism exists at all), a fixed settlement date (directly resolving the perpetual *gharar* objection), and smart-contract self-custody (no exchange counterparty risk). On-chain expiring futures have been implemented on protocols including MCDEX and Serum-based order books; their primary limitation is liquidity fragmentation across expiry dates rather than any structural Shariah concern. As on-chain liquidity infrastructure matures — particularly through unified expiry-date pools and cross-margining mechanisms — DEX expiring futures may offer the cleanest path to Shariah-compliant leveraged derivatives that also satisfies the most conservative scholarly positions on *gharar*. Dedicated empirical and jurisprudential analysis of these instruments is the most important near-term extension of this research agenda.

9 Conclusion

The Islamic finance consensus that perpetual futures are prohibited is correct — for CEX perpetuals. The error committed by 38 reviewed sources is an error of extrapolation: correct diagnosis of *riba* in one implementation was incorrectly generalised to all implementations. That generalisation was defensible as long as all implementations shared the same formula. It is no longer defensible.

This paper establishes four findings that together constitute a new paradigm.

First, the interest parameter I in CEX funding formulas is demonstrable *riba* by any classical definition — predetermined, fixed, labeled as interest, and unavoidable. Conservative scholars reached this verdict correctly, including SeekersGuidance (2025), which provides the most rigorous contemporary ruling. That verdict stands for all platforms with $I > 0$, whether CEX (Binance, Bybit, OKX) or DEX (Hyperliquid).

Second, the interest parameter is mathematically unnecessary. Ackerer, Hugonnier & Jermann (2024) proved this formally. DEX protocols dYdX v4 and D8X proved it practically — they have operated successfully with $I = 0$ at scale. Riba was chosen in 2016; it was never required.

Third, 365 days and 40,218 funding intervals across a natural experiment confirm that the interest parameter — not exchange architecture — determines distributional structure. Hyperliquid (DEX, $I > 0$) behaves like CEX. dYdX v4 (DEX, $I = 0$) behaves differently. The Shariah significance of this result is not statistical: it is the empirical proof that removing the interest parameter changes the observable character of the instrument in exactly the way Shariah analysis predicts.

Fourth, principles-based Shariah analysis — apply-

ing El-Gamal's riba conditions, Kamali's standardisation criterion, Sole & Jobst's four-part framework, and Salamon's maysir test — yields a coherent verdict: premium-only DEX perpetuals used on productive underlying assets with disciplined risk management do not violate the three canonical Islamic prohibitions. The *riba* is absent by design. The *gharar* is substantially reduced by on-chain transparency. The *maysir* determination is use-contingent, as it is for all derivatives.

The practical implication is significant. 1.8 billion Muslims face structural barriers to participation in the dominant leveraged instrument of modern crypto markets. This paper demonstrates that those barriers are not inherent to the instrument class — they are a consequence of specific funding formula design choices. Where the interest parameter is set to zero, as in dYdX v4 and D8X, the structural basis for prohibition is removed. This is a replicable design criterion, not a platform-specific accident: any perpetual futures exchange that adopts a premium-only funding formula satisfies the same structural condition.

The reform implication is therefore broader than a verdict on existing platforms. It is a specification: perpetual futures can be structured for Shariah compliance by design, and the empirical evidence presented here demonstrates that such structures function at institutional scale across a full market cycle. We call on Islamic scholars, Shariah boards, and standard-setting bodies — particularly AAOIFI and OIC Fiqh Academy — to engage with the technical specifics of DEX funding formulas and to develop formal guidance that enables institutional participants to evaluate perpetual futures instruments on a principled, formula-specific basis. The analytical framework is established; the empirical validation is documented; the formal *fatwa* must follow.

And Allah knows best.

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